



Python for Data Science - 2305CS303

Lab - 7

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}

```
In [4]: A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
print("Union:", A|B)
print("Intersection:", A&B)
print("Difference:", A-B)
```

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference: {1, 2}

02) WAP to Find elements common to all three given sets.

Sample Input:

A = {1, 2, 3, 4}
B = {2, 3, 5}

C = {2, 3, 6}

Sample Output:

{2, 3}

```
In [5]: A = {1, 2, 3, 4}
        B = {2, 3, 5}
        C = {2, 3, 6}
        print(A&B&C)
```

{2, 3}

03) WAP to remove duplicate elements from a list while preserving unique values.**Sample Input:**

[10, 20, 10, 30, 20, 40]

Sample Output:

{10, 20, 30, 40}

```
In [6]: l=[10, 20, 10, 30, 20, 40]
        s=set(l)
        s
```

Out[6]: {10, 20, 30, 40}

04) WAP to count how many elements are common between two sets.**Sample Input:**

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

2

```
In [7]: A = {1, 2, 3, 4}
        B = {3, 4, 5, 6}
        print(len(A&B))
```

2

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

```
In [15]: Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
for k,v in Dict2.items():
    if k in Dict1:
        Dict1[k]+=v
    else:
        Dict1[k]=v
Dict1
```

```
Out[15]: {1: 10, 2: 60, 3: 80, 4: 60}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

```
In [34]: subjects={'Math': 70, 'Physics': 85, 'Chemistry': 60}
print(dict(sorted(subjects.items(),key=(lambda t:t[1]))))
```

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.**Sample Input:**

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

```
In [44]: d={'A': 50, 'B': 90, 'C': 30}
print("Max Value key: ",max(d.items(),key=(lambda t:t[1]))[0])
print("Min Value key: ",min(d.items(),key=(lambda t:t[1]))[0])
```

Max Value key: B

Min Value key: C

```
In [81]: d={'A': 50, 'B': 90, 'C': 30}
d[max(d,key=lambda k:d[k])]
```

Out[81]: 90

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

N = 5

Sample Output:

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

```
In [58]: n=5
d={k+1:v**2 for k,v in enumerate(range(1,n+1))}
d
```

Out[58]: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

programming

Sample Output:

{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}

```
In [62]: s="programming"
d={}
for i in s:
    if i not in d:
        d[i]=1
    else:
        d[i]+=1
d
```

Out[62]: {'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

```
In [63]: d={1: 'A', 2: 'B', 3: 'C'}
r={}
for k,v in d.items():
    r[v]=k
r
```

```
Out[63]: {'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.**Sample Input:**

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

```
In [69]: d={'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
print("Topper: ",max(d.items(),key=(lambda t:sum(t[1])))[0])
```

```
Topper: Amit
```

12) WAP to extract keys whose values are even numbers.**Sample Input:**

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

```
In [70]: d={1: 11, 2: 20, 3: 15, 4: 40}
e=[v for k,v in d.items() if v%2==0]
e
```

```
Out[70]: [20, 40]
```

13) WAP to Find common keys between two dictionaries.**Sample Input:**

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}  
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

```
In [77]: Dict1 = {1: 'A', 2: 'B', 3: 'C'}  
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}  
result=[k for k,v in Dict1.items() if k in Dict2]  
result
```

```
Out[77]: [2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}  
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}  
key = 'd'
```

Sample Output - 2:

```
Key Not Found
```

```
In [84]: dict1 = {'a': 5, 'c': 8, 'e': 2}  
key='c'  
if key in dict1:  
    print(dict1[key])  
else:  
    print("Key Not Found")
```

```
8
```

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

```
In [91]: students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
res={k:v for k,v in zip(students,marks) if v >=75}
res
```

```
Out[91]: {'Amit': 78, 'Neha': 85}
```

16) d = {"GFG": [5,7,5,4,5],"is":[6,7,4,3,3],"Best":[9,9,6,5,5]}

OUTPUT

'is'

```
In [106... d = {"GFG": [5,7,5,4,5],"is":[6,7,4,3,3],"Best":[9,9,6,5,5]}
r=[max(d.items(),key=lambda t: len(set(t[1])))[0]]
r
```

```
Out[106... ['is']
```

17) Take string and replace word from dictionary

```
In [119... s="hello world, have a great day"
d={"hello":"hi","world":"earth","great":"wonderful"}
for i in d:
    s=s.replace(i,d[i])
s
```

```
Out[119... 'hi earth, have a wonderful day'
```

18) remove duplicate from string after first occurrence

```
In [123... s="Hello students how are you students how is day going".split()
l=[]

for i in s:
    if i not in l:
        l.append(i)
print(" ".join(l))
```

Hello students how are you is day going